

EEE 1101: Electrical Circuits

He who knows not, and knows not that he knows not, is a fool— shun him.

He who knows not, and knows that he knows not, is a child— teach him.

He who knows, and knows not that he knows, is a sleep—wake him up.

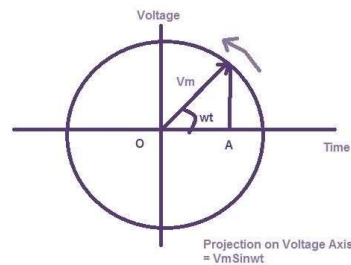
He who knows, and knows that he knows, is wise—follow him.

—Persian Proverb

Vector and phasor:

Vector: A vector is a quantity that has both magnitude and direction.

Phasor: A phasor is a rotating vector that represents the magnitude and phase of a sinusoidal waveform at a constant angular frequency.



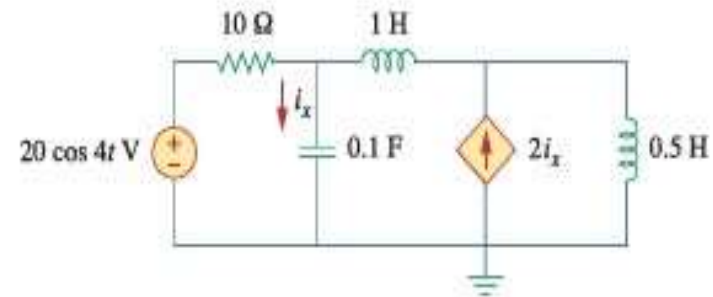
Every **phasor** is a kind of **vector**, but not every **vector** is a **phasor**.

Instantaneous Value: The **instantaneous value** is the **actual value** of a waveform at any **specific moment** in time.

Maximum Value: The **maximum value** (also called **peak value**) of an alternating quantity is the **highest value** it reaches during one cycle.

RMS Value: The **RMS (Root Mean Square) value** of an alternating quantity is the **effective value** of that quantity which produces the **same heating effect** in a resistor as an equivalent **DC value**.

Q. Determine i_x using nodal analysis.

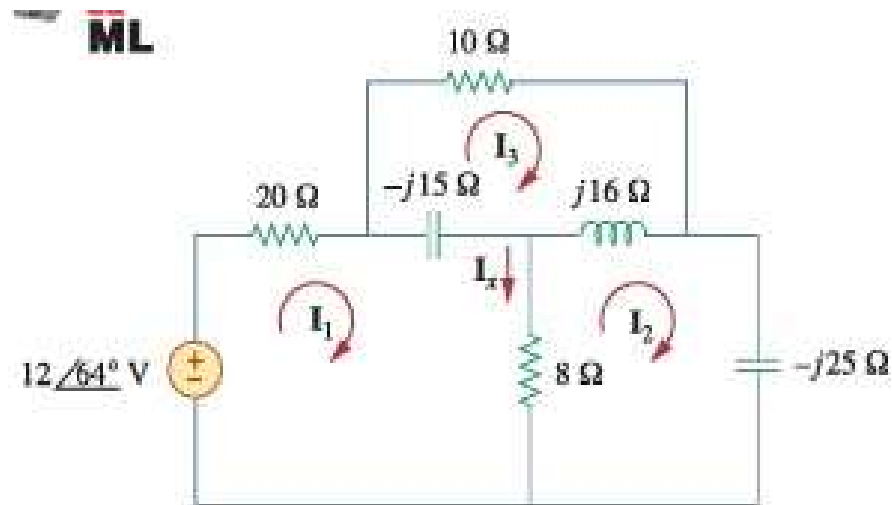


$$\mathbf{V}_1 = \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97 \angle 18.43^\circ \text{ V}$$

$$\mathbf{V}_2 = \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91 \angle 198.3^\circ \text{ V}$$

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

Q. Find I_1 , I_2 , I_3 and I_x using Mesh analysis.

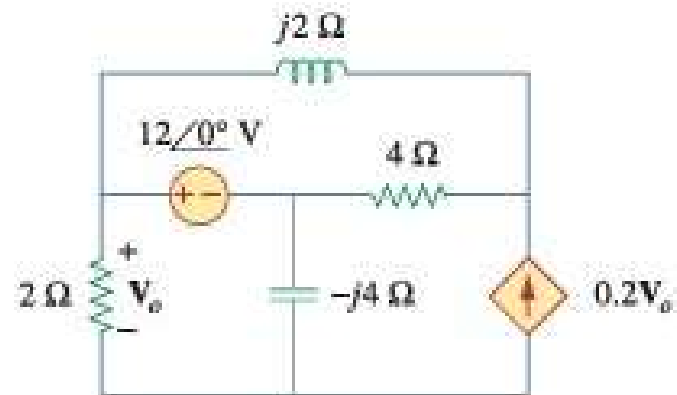


381.4/109.6° mA, 344.3/124.4° mA,
145.5/-60.42° mA, 100.5/48.5° mA

EQ1. Determine V_x using nodal analysis.



EQ2. Determine V_o using nodal analysis. (10.19)



EQ3. Find V_o and I_o using Mesh analysis. (10.32)



EQ4. Find I_o using Mesh analysis. (10.38)

